Explainable Fake News Detection with Transformers

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Fake news is everywhere. We propose a system that uses RoBERTa + progressive fine-tuning to tell real from fake. Also we add LIME so you can see why the model thinks something is fake. Our best model reaches 94.7% F1 (hopefully – we'll report real numbers).

Introduction

Social media is full of lies. But most detectors are black boxes – you don't know why they say "fake". That's annoying if you're a journalist or a fact-checker. So we decided to build something that shows its work.

Problem Statement

Given a news article text, output: FAKE or REAL + a highlight of suspicious words.

Why this matters

Because misinformation kills (literally, during COVID). Also the professor likes explainable AI.

Related Work (brief)

BERT did well on LIAR (Wang, 2017). RoBERTa did better (Liu et al., 2019). LIME (Ribeiro et al., 2016) explains predictions. But nobody combined progressive fine-tuning + LIME for fake news – or if they did, we didn't find it.

Proposed Approach

We'll compare:

- TF-IDF + logistic regression (boring baseline)
- DistilBERT (fast)
- RoBERTa (accurate)

- RoBERTa with progressive fine-tuning (first on Kaggle data, then on LIAR2)

Then we'll explain predictions with LIME.

Datasets

LIAR2 (23k statements) and Kaggle fake-real news (40k articles). We binarize LIAR2's 6 labels into fake (0-2) and real (3-5).

Work Breakdown (who does what)

Salah Eddine	Data, baseline, paper writing
Oussama	RoBERTa, progressive fine-tuning
Mohamed	Metrics, LIME, error analysis
Nadjemeddine	Streamlit demo, slides

Timeline (optimistic)

Week 1: Proposal Week 2: Data prep Week 3: Baseline + DistilBERT Week 4: RoBERTa + progressive Week 5: Evaluation + LIME Week 6: Demo + paper draft Week 7: Final paper Week 8: Presentation

Expected Results

Accuracy > 90% on LIAR2. F1 > 0.9. LIME will highlight words like "hoax", "conspiracy", "vaccine".